

CFG PRACTICE SHEET

AUTOMATA & COMPUTABILITY (CSE331)

[Give a context-free grammar for each of the following languages.]

1. $\{w \in \{0,1\}^*: 0^n 1^n \text{ or } 1^n 0^n \text{ where } n \geq 0\}$
2. $\{\text{starts and ends with different symbols}\}$
3. $\{\text{starts and ends with same symbols}\}$
4. $\{\text{has exactly two 1's}\}$ [Summer23]
5. $\{w \text{ is a palindrome}\}$
6. $\{w \text{ is not a palindrome}\}$ [Spring24]
7. $\{w \text{ is a valid parenthesis}\}$
8. $\{w \in \{0,1,2\}^*: 0^i 2^j 1^k \text{ where } i=k, i,j,k \geq 0\}$
9. $\{w \in \{a,b,c\}^*: a^i b^j c^k \text{ where } i+j=k, i,j,k \geq 0\}$
10. $\{w \in \{a,b,c\}^*: a^i b^j c^k \text{ where } i=j \text{ or } j=k, i, j, k \geq 0\}$
11. $\{w \text{ is even length}\}$
12. $\{w \text{ is odd length}\}$
13. $\{w \in \{0,1\}^*: 1^n \text{ where } n \geq 0\}$
14. $\{w \in \{0,1\}^*: 0^i 1^j 0^k \text{ where } j = i+k, i, k \geq 0\}$
15. $\{w \in \{0,1\}^*: 0^n 1^{2n} \text{ where } n \geq 0\}$
16. $\{w \in \{0,1\}^*: 0^{3n} 1^{5n} \text{ where } n \geq 0\}$
17. $\{w \in \{0,1\}^*: 0^{n+1} 1^n \text{ where } n \geq 0\}$
18. $\{w \in \{0,1,2\}^*: 0^n 12^n \text{ where } n \geq 0\}$
19. $\{w \in \{0,1,2\}^*: 0^n 1^n 2 \text{ where } n \geq 0\}$
20. $\{w \in \{0,1\}^*: 0^n 1^{n+1} \text{ where } n \geq 0\}$
21. $\{w \in \{0,1\}^*: 0^m 1^m \text{ where } m \geq 2\}$
22. $\{w \in \{0,1\}^*: 0^m 1^n \text{ where } m \geq n\}$
23. $\{w \in \{0,1,2,3\}^*: 0^m 1^n 2^k 3^j \text{ where } k = m + n + j\}$
24. $\{w \text{ contains at least 2 0's}\}$ [Spring22]
25. $\{\text{length odd in middle 0}\}$
26. $\{w \in \{0,1,2\}^*: 0^i 2^j 1^k \text{ where } i=j \text{ or } j \neq k, i,j,k \geq 0\}$
27. $\{w \text{ contains even number of 0's}\}$
28. $\{w \text{ contains twice as many 1s as 0s}\}$
29. $\{w \text{ contains even number of 0s and 1s}\}$
30. $\{w \text{ where each 0's is followed by at least as many 1's}\}$
31. $\{w \in \{a,b\}^*: a^n b^m \text{ where } 0 < n < m < 3n\}$
32. $\{w \text{ contains at least 4 a's}\}$
33. $\{w \in \{a,b,c,d\}^*: a^n b^m c^m d^{2n} \text{ where } n \geq 0, m > 0\}$
34. $\{w \in \{0,1\}^*: \text{no of 0's and 1's in } w \text{ is same}\}$
35. $\{w \in \{0,1\}^*: \text{the parity of 0's and 1's are different}\}$
36. $\{w \in \{0,1\}^*: \text{no of 0's and 1's in } w \text{ is different}\}$
37. $\{w \text{ starts with 0 and length of } w \text{ is even}\}$ [Spring24 only 2nd part]
38. $\{\text{every second letter in } w \text{ is } b\}$
39. $\{\text{the length of } w \text{ is divisible by 3}\}$ [Spring24]
40. $\{\text{the length of } w \text{ is more than multiple of 5}\}$
41. $\{w \in \{0,1\}^*: 0^n 1^{2n+3} \text{ where } n \geq 0\}$
42. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } j \geq 2i + 3k \text{ and } i, k \geq 0\}$
43. $\{w \in \{0,1,2,3\}^*: 0^i 1^j 2^k 3^m \text{ where } i=m, j \geq 3k+2 \text{ and } m, k \geq 0\}$
44. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } i > 2j + 3k \text{ and } j, k \geq 0\}$
45. $\{w = 0^{3i} v 2^{2k}, v \in \{\text{at least two 0's}\} \text{ and } i \geq 0\}$ [Spring22]
46. $\{w \in \{a,b\}^*: w \text{ starts with even no of a's}\}$
47. $\{w \in \{a,b\}^*: w \text{ starts with odd no of b's}\}$
48. $\{w \in \{0,1,2\}^*: 1^i 02^j 1^k \text{ where } i=k, i,j,k \geq 0\}$ [Fall22]
49. $\{w \in \{0,1,2\}^*: 1^i 02^j 1^k \text{ where } k=i+2j, i,j,k \geq 0\}$ [Fall22]
50. $\{w \in \{0,1,2\}^*: a^i \# b^j \text{ where } i \text{ is a multiple of three, } i,j,k \geq 0\}$ [spring23]

51. $\{w \in \{0,1,2\}^*: a^i \# b^j \mid j=2+3i, i,j,k \geq 0\}$ [spring23]

[Convert the following Regular expressions to a CFG]

1. $a(b|c)^*$
2. $0^*1(0+1)^*$
3. $(a+b)^*(a^*+(ba)^*)$
4. $(a+b)^*aa(a+b)^*$
5. $a^*+a(a|b)^*$
6. $0(0+1)^*1+1(0+1)^*0$
7. $(ab+bc)bb(ca+cc)^*+(aa)^*$
8. $a^*(ba+bb)^*+\#$ [Fall23]
9. $a(ba)^*+\#a^*b$ [Fall23]

[Give leftmost and rightmost derivations for the following strings and check parse-tree ambiguity.]

1. Give a leftmost derivation for string w .
2. Sketch the parse tree for the (1) leftmost derivation (w) string.
3. If it is ambiguous or not?
4. Make a six length string which has only one parse tree.

a. $S \rightarrow 0A \mid 1B$
 $A \rightarrow 0AA \mid 1S \mid 1$
 $B \rightarrow 0S \mid 1BB \mid 0$
String: 001101

b. $S \rightarrow A \mid B$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1B \mid \epsilon$
Strings: 10100, 0010101

b. $S \rightarrow A \mid B$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1B \mid \epsilon$
Strings: 10100, 0010101

c. $D \rightarrow TL$
 $T \rightarrow c \mid Tc$
 $L \rightarrow L.V \mid V$
 $V \rightarrow a \mid b \mid 0 \mid 1 \mid Va \mid Vb \mid V0 \mid V1$
Strings: cabb0011.ab1 (Rightmost derivation)

d. $S \rightarrow S + S \mid Sa \mid Sb \mid \epsilon$
String: abb + aab + baba

e. $S \rightarrow SA \mid \epsilon$
 $A \rightarrow aa \mid ab \mid ba \mid bb$
String: aabbba

f. $S \rightarrow aEbS$
 $S \rightarrow aEbScS \mid \epsilon$
 $E \rightarrow d$
String: adbadbadbc

g. 00010111
Where $S \rightarrow ASB \mid SS \mid SAS \mid A$
 $A \rightarrow ASS \mid BS \mid B$
 $B \rightarrow 00 \mid 11 \mid 01 \mid 10$

h. 001111
Where $A \rightarrow A1 \mid 0A1 \mid 01$